Technology Mapping over E-Graphs with Delay-Constrained Area Recovery

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***Abstract*—Technology mapping must select a realizable circuit from equivalent Boolean structures while accounting for shared logic and timing. We present eGraphMap, a prototype for single-output standard-cell mapping that enumerates cuts directly over an e-graph and represents matches as physical instances. Whole-cover metrics charge reused instances once and account for input phase and pin-specific delay. Exact identities remain inside the e-graph; conservative satisfiability don't-cares admit additional external matches without introducing conditional equalities globally. Extraction combines a finite acyclic witness, bounded recovery, and an optional mixed-integer formulation. Experiments use all 20 EPFL arithmetic and control circuits, two restricted ASAP7 inventories, one shared `resyn2` front end, and original baseline algorithms. A complete 120-case proportional-capacity study compares three search ratios. The single selected ratio produces 40 verified covers across 40 circuit/library cases, of which 38 select an optimized e-graph cover and 2 retain a basic source implementation. Targeted repetitions retain failures and area or timing regressions. The new capacity policy removes fixed-size import rejection but does not eliminate subsequent search timeouts or establish full-suite dominance.**

***Index Terms*—technology mapping, equality saturation, shared logic, area recovery, satisfiability don't-cares**

I. Introduction

The quality of a mapped circuit depends on both the available Boolean structures and how their implementations share physical cells. A rewrite that looks attractive as a tree may duplicate a shared cone. Even with single-output cells, independent choices at different roots can lose sharing across reconvergent paths. Reachability provides another opportunity: assignments to a cut boundary that can never occur permit matches that fail a fully specified truth-table comparison.

Equality saturation addresses structural commitment by recording many equivalent expressions in an e-graph [1]. Recent synthesis work has already used this representation for technology-aware rewriting, structural exploration, and FPGA LUT remapping [4], [5], [6]. Multi-output standard-cell mapping also has established algorithms [2]. Combining these techniques requires a mapping model that distinguishes Boolean equivalence from physical sharing and from conditional matching. We focus on single-output mapping, with multi-output cells retained as an extension.

The study combines three mechanisms. First, structural cuts and exact functional-quotient boundaries are constructed over the retained e-graph, rather than committing to one extracted Boolean network. Second, matching retains input phase, pin-specific delay and conditional care information; SDC matches remain outside the global equivalence relation. Third, whole-cover recovery separates physical-instance activation from binding, charges shared cells once, and protects an already visited seed while reducing actual area under a validated delay bound. Explicit source proofs accompany exchanged bindings. The implementation exports acyclic netlists and checks primary-output equivalence independently. Joint-output selection is retained as an extension.

Standard-cell evaluation uses single-output ASAP7 subsets; synthetic libraries test multi-output mechanisms separately. All methods share the preoptimized input and netlist evaluator. Synthetic cell advantages are not attributed to the standard-cell results.

II. Background and Related Work

A. Mapping and shared logic

Cut-based mapping summarizes the function at a root in terms of a bounded set of boundary signals. Matching assigns that function to a library cell under input permutation and phase choices. DAG-aware area-flow and exact local area recovery account for sharing more effectively than additive tree costs. The versatile mapping approach implemented in mockturtle provides an important standard-cell baseline [3].

Calvino and De Micheli address multi-output matching and whole-network optimization [2]. We retain e-graph alternatives and conditional matches during candidate generation; emap receives the same library/input.

B. Equality saturation in synthesis

E-Syn introduces technology-aware costs for Boolean e-graph rewriting [4]. E-morphic scales structural exploration before mapping using pruning and extraction techniques [5]. We evaluate both methods after a common front end. E-Syn uses a bounded search with frozen cost models; E-morphic uses its final simulated-annealing extractor. Their library restrictions and evaluation settings are specified in Section V.

EqMap uses e-graphs for FPGA LUT remapping [6], optimizing LUT count and depth. We evaluate a cross-target composition by flattening its LUT network and applying a standard-cell backend. A control with the same LUT conversion and backend isolates the effect of remapping. These results measure the composition rather than native FPGA resource utilization.

Churchroad combines equality saturation with program synthesis for target-specific FPGA primitives [7]. Its DSP-oriented target differs from the standard-cell libraries considered here.

III. Problem Formulation

Let G be a combinational Boolean network with primary inputs P and primary outputs R, and let $\mathcal{L}$ be a library. A physical library cell g has area a_g, a common input-pin set I_g, and output-pin functions $f_{g,p}$. Its delay model contains input-to-output block delays $d_{g,p,j}$ for functionally dependent inputs j. We seek a finite acyclic implementation equivalent to G on every primary input assignment. We compare two lexicographic objectives:

$$\min_{\text{lex}}(A, D) \quad \text{or} \quad \min_{\text{lex}}(D, A), \qquad A = \sum_{m \in \mathcal{M}} a_m x_m, \quad (1)$$

optionally subject to $D \leq D_{\max}$. Here x_m selects a physical candidate instance. Area and delay use the supplied genlib units. The primary metric is minimized first; the secondary metric breaks ties within numerical tolerance. No external target clock is imposed; internal area recovery instead fixes the delay of its validated reference cover. For integer extraction, a second solve constrains the best known primary value and minimizes the secondary metric within the remaining time budget. Lexicographic optimality is certified only if both stages prove optimality and their extracted covers pass validation; a timeout preserves the best checked cover without that certificate. Heuristic acceptance uses the same priority order, but does not guarantee optimality. A lexicographic win is reported separately from simultaneous Pareto improvement.

Each e-class denotes an exact Boolean equivalence class. A signal combines an e-class identifier and a phase bit, permitting explicit inverted signals and materialized inverter costs. A cut $C = (B, F)$ has an ordered boundary B and a truth table F with $|B| \leq K$. Candidate m specifies a cell, a common assignment of physical inputs to boundary signals, and one or more admissible bindings (m, p, s) between output pin p and target signal s. Multiple targets of the same pin share a physical net when selected. Output pins may remain unbound.

IV. Proposed Method

A. Bounded cuts over exact alternatives

The importer hash-conses Boolean operators into the e-graph. The proportional profile uniformly enables exact XOR and three-input truth-table cone compression, independently of capacity. All operands and admitted helpers count globally; the mandatory helper quota is 2048 and optional fill is disabled. Source-exact physical cuts use the existing cut banks. This changes the node unit relative to expanded ANDs; original source counts remain recorded. Expression positions use returned e-class identifiers. Exact rewrites preserve Boolean semantics. Iteration and node limits bound saturation, after which canonical e-classes are used to enumerate cuts across alternative e-nodes.

Cut construction merges child boundaries and composes their truth tables. Structural boundary cuts are retained to help preserve a finite original implementation. Dynamic ranking limits alternatives per e-class; cyclic e-graph structure does not imply that a cyclic implementation is allowed. The implementation separately constructs and validates a finite witness. The configured budgets bound exploration, so extraction optimality, when reported, applies only to the finite candidate problem. Region ownership preserves complete physical boundaries and cross-region targets in one global e-graph. A revised admission ledger charges PIs, operators, helpers and rewrite additions before insertion. Union and rebuild never refund its capacity; regions do not reset it. Opaque tables can hide rewrite patterns. Atomic producer/inverter changes validate the global cover; a phase window reserves local recovery time under the same deadline.

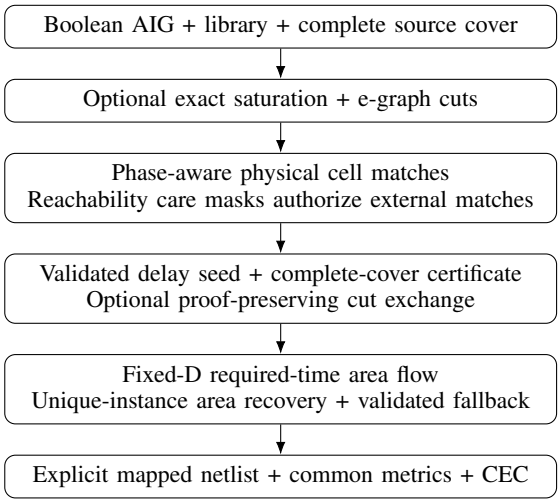


Fig. 1. Implementation flow. Conditional matches do not add equalities to the exact e-graph.

An optional retention profile reserves eight small-support cuts within the same per-class allowance and interleaves intrinsic-delay and witness-arrival producer ranks. Exact Not edges identify physical signal aliases; raw e-class cut banks and physical pin interfaces remain intact. Aliases are merged before physical-instance admission. Required-time area recovery uses mapped references and validates complete covers. Conditional matches establish no aliases. Temporary generation can exceed the retained-pool cap; all recovery uses the original deadline.

Exploratory functional cuts form $F(S) = Q(S \setminus T, G(T))$ using an existing class for $G(T)$, $T \subset S$. Full assignments must uniquely determine every quotient entry. Up to four partners per support and 32 trials per class per original round compete in the same 16 slots, adding no e-nodes or conditional unions; old cuts may be displaced.

B. Physical cell matching

For each cut function, matching enumerates legal input permutations and phases using the cell truth table and its physical pin delays. The physical interface is kept explicit: two assignments with equal Boolean functions can have different delay arcs. A single-output instance can serve several equivalent targets or fanouts, but its area is charged only

once. Functionally independent pins contribute connectivity requirements without contributing timing arcs. Each admitted binding retains its generating boundary, care mask, target phase and physical pin assignment. Deduplication merges proofs rather than discarding them. These records are local to one mapping invocation; conditional care never licenses an e-class union.

Joint-output cells are an extension of this representation. Records with the same physical cell name share an input assignment and one area charge, while retaining separate output functions and delay vectors. Bounded unions of neighboring or shared-leaf boundaries admit different output supports; they do not exhaust all possible boundary unions. Full-adder and disjoint-support synthetic tests exercise joint selection, open unused pins and shared-instance accounting. The realistic inventories evaluated here contain no multi-output cells.

C. Conservative satisfiability don't-cares

For a boundary $B = (b_1, \ldots, b_k)$, define the reachable image

$$\mathcal{R}_B = \{(b_1(u), \ldots, b_k(u)) : u \in \{0,1\}^{|P|}\}. \tag{2}$$

A candidate output function H may bind to target F if

$$(F \oplus H) \wedge C_B = 0, \qquad \mathcal{R}_B \subseteq C_B, \tag{3}$$

where C_B is interpreted as a bit mask of care assignments. The prototype traces the original boundary cones and enumerates their combined primary input support only when both support and cone budgets permit. This yields the exact reachable image for those bounded cases. Otherwise it uses the full care set, losing opportunities rather than admitting unsound matches.

Matches remain external: agreement on reachable values is not a generic identity for independent boundary variables. For fixed $B(X)$, Equation (3) proves $F(B(X)) = H(B(X))$; a concrete proof-backed union could be sound, but we choose external storage. We handle SDCs, not ODCs, environmental assumptions or sequential reachable states.

D. Acyclic cover extraction

For each admissible output binding (m, p, s) introduce a binary variable $z_{m,p,s}$. Let $\mathcal{B}(s)$ be its producer bindings and define $q_s = \sum_{(m,p,s)\in\mathcal{B}(s)} z_{m,p,s}$. Non-free signals satisfy $0 \le q_s \le 1$, required roots satisfy $q_s = 1$, and every binding obeys $z_{m,p,s} \le x_m$. A selected instance requires availability of every non-free physical input j, with free signal set $\mathcal{F}$:

$$x_m \le q_j, \qquad j \in I_m \setminus \mathcal{F}. \tag{4}$$

Constants and primary input signals are free. Complemented inputs require their actual producers unless an explicit library match realizes them.

Continuous levels ℓ_s prevent cycles. For every bound output and every common physical input,

$$\ell_s \ge \ell_j + 1 - M_\ell(1 - z_{m,p,s}). \tag{5}$$

These constraints apply to secondary output pins as well as the first. Requiring all physical inputs is conservative when a pin is irrelevant to the selected output, but ensures a realizable connected instance.

When timing participates in extraction, arrival variables t_s satisfy

$$t_s \ge t_j + d_{m,p,j} - M_t(1 - z_{m,p,s}) \tag{6}$$

for functionally dependent inputs only. Primary input arrival is zero and $D \ge t_r$ for each root r. The implementation bounds levels and arrivals before choosing the big-M constants. Independent metric recomputation rejects cyclic, incomplete, fractional, or timing-infeasible solutions; solver termination alone is insufficient.

E. Recovery and safe fallback

A validated finite witness initializes recovery. A dependency frontier releases producers with available inputs and supplies an acyclic delay cover. A witness already required for candidate retention may cross this interface; index remapping and validation consume the same recovery window. Reporting-only frontiers cannot seed the mapper.

Before exchanging cuts, we preserve the complete visited cover, including noncritical shared dependencies. Its bindings must retain generator proofs in the new bank. A borrowed preparation view copies only changed records, with a 2048-record limit. The commit moves unchanged records. Equal physical inputs, pin arcs and area, together with injective producer renaming, preserve the saved DAG and metrics. Failed preparation leaves the old pool untouched; disposal consumes the same global window. Unfinished support indices are rejected rather than used partially.

Area flow propagates required times from a fixed validated delay limit. Actual-area recovery charges unique physical instances and refreshes labels after accepted changes. A separate archive retains eligible, already-scored whole covers under delay-area order: an over-area fast point cannot suppress a feasible intermediate. Final validation enforces both seed limits. This extra area constraint neither reproduces another pipeline's historical final cover nor proves global exact-area optimality. A timeout carries no optimality certificate.

F. Correctness scope

Assume exact rewrites, truthful cell functions and care masks containing the original reachable image. Topological induction over an acyclic cover establishes each bound signal's original semantics using Equation (3), hence output equivalence. This does not justify independent ODC replacements or unproved generic conditional identities. Explicit cell models, Yosys flattening and named-port ABC CEC [8] independently check emitted networks. Unknown outcomes, missing tools, undriven nets and cycles fail; corrupted-output tests exercise rejection.

V. Experimental Evaluation

A. Protocol

The suite contains all 20 EPFL arithmetic/control circuits [10]; MtM is not included. Each circuit uses one named AIG

produced by a fixed ABC `resyn2` sequence and independently checked against the original input. Both inventories and every method reuse this AIG. The three-input and four-input inventories are restricted subsets of the vendored ASAP7 genlib [9], not the full PDK.

Conventional baselines are ABC `map -a`, ABC `map`, and independent emap in area and delay modes. Equality-saturation comparisons use EqMap, E-Syn, and E-morphic. Their original algorithms, models, and native parameters are retained. Necessary indexed-port adaptation and mapping-clock instrumentation are documented separately. No repaired E-Syn feature/index probe is used as a baseline.

Each circuit/method case has a cumulative 300-second mapping allowance; preparation, conversion, CEC and metric reporting are excluded. Original process guards are 60 seconds, or 90 seconds for EqMap. Native budgets differ. E-morphic feedback time is the union of active mapping intervals. EqMap's initial nested LUT stage uses enclosing Yosys elapsed time as a conservative budget upper bound; its exact mapping runtime is unavailable. Mapper process guards include internal reports. Full optional candidate diagnostics are disabled uniformly; exact counts, admission records and recovery traces remain available; Tabulated T excludes external preparation, CEC and evaluation. Bounds are rounded up to 0.01 seconds and include EqMap's enclosing LUT stage or eGraphMap internal reports; no isolated algorithmic speedup is inferred.

The fixed eGraphMap profile uses delay-area priority, 256-class region ownership, dependency-driven flow and a 0.25 phase window. Stage checks occur between atomic trials; the fractions are not preemptive caps. Small-cut quota8, mixed retention, exact physical aliases, mapped references and required-time area recovery are enabled. Capacity is determined before mapping from the common input AIG:

$$B = 2(|P| + |\mathrm{AND}|) + 2, \qquad G = B + \lceil \rho B \rceil. \tag{7}$$

The three fixed ratios are $1/2$, 1 and 2. PIs, operators, helpers and rewrite additions share the cumulative ledger; rebuild and retirement refund nothing. Unused input reservation remains available in this ledger, rather than constituting a separate saturation counter. This is a new resource protocol, not an unchanged 20,000-node comparison.

A direct AND/NAND/INV source cover is constructed before optional search. The native retained bank remains limited to 100,000 candidates; the separately retained complete source incumbent has a conservative global bound of $B + 100{,}000$. Retained counts do not bound transient generation memory. The other limits remain 16 cuts/class, 16 saturation iterations, 32 cut rounds and four recovery rounds of 2,000 local trials. Synchronous dependency notifications avoid unchanged cut recomputation; streamed physical normalization removes duplicates on emission. Root-cover retention protects recursive dependencies of the complete output witness, without reserving witnesses for every unselected internal signal.

The global two-second recovery window includes mandatory post-cap availability reconstruction, index remapping and final validation. A 0.25-second return reserve and the existing 0.25 phase fraction remain inside that window. Earlier retention ranking remains within the original native guard, outside recovery. CBC is disabled. The search profile retains functional cuts, phase/pin matching, SDC and fixed-bound shared-area recovery. Final export compares complete source and optimized covers under delay-area priority; source fallback is not counted as optimization success.

The evaluator charges all physical instances once, including buffers. Delay is the maximum output arrival under maximum rise/fall block delay for functionally dependent pins; load, slew and interconnect are omitted. Every numeric record has named-port CEC against the original AIG. Fresh optimized outputs use independent functional Verilog cell models, with Boolean normalization confined to verification after physical metrics are measured. Exact source-netlist/input/library/evaluator hashes permit reuse of an existing CEC proof; reused proofs are not fresh CEC runs. Verification has a separate 600-second guard and never supplies candidates or scores to the mapper. Seven of eight verification-only baseline supplements succeed; their original mapping netlists, times, algorithms and failed reports remain unchanged. Manifests retain source/input/library/binary hashes, settings, commands, netlists and failed logs. Script-file CEC avoids the OS argument-length limit without changing functions or port matching.

B. Coverage and standard-cell results

All 240 original baseline groups were attempted. With seven additional verification passes, their 520 method/status slots contain 379 CEC passes, 117 reported failures and 24 unmeasured slots. An eighth verification supplement still times out; no mapping is rerun or baseline optimized. Commands distinguish conversion guards, mapping timeouts, native exits and verification failures. Attempted coverage is not valid PPA coverage.

All 120 proportional cases are retained in Table IV. The selected global ratio is $\rho = 1/2$, shared by both inventories and all 20 designs. Selection first maximizes verified coverage, then minimizes common-scope geometric delay, area and finally the ratio, using relative 10^{-6} ties. This is an in-suite configuration study, not evidence of out-of-suite generalization. Actual admissions are identical across ratios in all 40 cases; no node cap is exhausted. Small area differences do not establish a repeatable benefit from extra capacity.

Tables I and II retain horizontal A/D/T columns for ABC map, delay-mode emap, depth-oriented EqMap, delay-oriented E-Syn, native final E-morphic SA and this one eGraphMap configuration. Only first full-sweep points enter these tables; repeated best points never replace them. Superscript s distinguishes basic source fallback from successful optimization. Area-oriented baselines, staging controls and every failure remain in the machine-readable comparison. Full-scope delay-area and Pareto gates remain false.

TABLE I

EPFL 3-INPUT INVENTORY: PHYSICAL AREA (A), UNLOADED BLOCK DELAY (D), AND MAPPING RUNTIME IN SECONDS (T). $\leq$ DENOTES A CONSERVATIVE UPPER BOUND; – DENOTES UNAVAILABLE RUNTIME. ONE UNIFORM EGRAPHMAP PROFILE AT $\rho = 1/2$; s MARKS ITS BASIC SOURCE COVER, NOT SUCCESSFUL OPTIMIZATION. F RETAINS FAILURE STATUS.

Design	ABC map			emap			EqMap			E-Syn			E-morphic			eGraphMap		
	A	D	T	A	D	T	A	D	T	A	D	T	A	D	T	A	D	T
adder	80.71	2585.48	0.06	90.33	2585.46	0.03	222.33	7485.96	$\leq$3.01	F	F	–	91.47	3536.80	7.61	89.50	2585.46	$\leq$1.18
bar	241.27	190.87	0.10	238.65	190.87	0.05	214.59	453.18	$\leq$3.16	F	F	–	284.68	194.42	14.73	239.91	190.87	$\leq$3.11
div	3043.49	43602.20	1.40	2931.67	43594.09	0.93	F	F	–	F	F	–	F	F	–	4016.31	43594.08	$\leq$16.37
hyp	15069.91	233873.90	9.24	14349.98	233105.95	4.48	F	F	–	F	F	–	F	F	–	22414.78^s	667115.20	$\leq$65.98
log2	1947.57	4225.92	1.31	1865.96	4189.37	0.70	F	F	–	F	F	–	F	F	–	2248.43	4189.37	$\leq$13.77
max	220.97	2111.03	0.10	213.13	2111.03	0.06	382.23	5559.50	$\leq$6.30	F	F	–	212.86	1726.03	20.43	232.37	2111.03	$\leq$2.95
multiplier	1721.22	2688.12	0.95	1704.19	2678.12	0.51	F	F	–	F	F	–	F	F	–	2009.79	2805.50	$\leq$9.42
sin	385.06	2076.78	0.29	372.38	2049.52	0.13	839.10	4880.69	$\leq$10.85	F	F	–	F	F	–	453.38	2050.82	$\leq$6.71
sqrt	1675.89	60089.98	0.75	1644.87	60087.26	0.42	F	F	–	F	F	–	F	F	–	1586.26	60061.59	$\leq$9.85
square	1187.58	2540.25	0.68	1116.66	2535.25	0.35	3189.89	6550.05	$\leq$23.63	F	F	–	F	F	–	1430.14	2535.25	$\leq$9.21
arbiter	751.05	900.55	0.41	750.02	900.55	0.26	1152.43	2024.28	$\leq$19.39	F	F	–	F	F	–	762.62	900.55	$\leq$5.80
cavlc	41.65	197.59	0.06	41.57	197.59	0.02	77.92	606.30	$\leq$2.16	40.81	274.78	1.30	42.92	185.01	5.70	41.71	197.59	$\leq$0.68
ctrl	7.37	108.06	0.04	7.26	108.06	0.01	9.77	175.06	$\leq$1.62	7.39	147.40	1.01	8.11	91.71	2.89	7.15	108.06	$\leq$0.22
dec	27.68	82.92	0.05	27.68	82.92	0.02	46.64	145.54	$\leq$1.48	27.22	90.68	1.11	27.68	84.91	3.51	27.68	82.92	$\leq$0.33
i2c	74.34	166.78	0.07	73.28	166.78	0.03	105.69	452.49	$\leq$1.92	F	F	–	72.97	149.83	9.36	73.99	166.78	$\leq$2.04
int2float	13.73	183.34	0.04	13.50	183.34	0.01	21.81	326.81	$\leq$1.25	13.34	221.24	1.18	14.86	145.87	5.42	13.59	183.34	$\leq$0.27
mem_ctrl	2733.79	1231.54	1.22	2610.21	1231.54	0.82	F	F	–	F	F	–	F	F	–	3448.17	1228.25	$\leq$15.45
priority	68.57	2062.18	0.06	67.58	2062.18	0.03	56.95	3157.07	$\leq$3.00	49.19	2872.79	1.31	41.08	1167.56	15.68	69.47	2062.18	$\leq$0.83
router	13.35	216.40	0.05	15.07	216.40	0.01	11.91	625.62	$\leq$1.55	12.02	390.69	1.20	15.18	198.42	3.66	13.72	213.20	$\leq$0.24
voter	747.46	826.69	0.43	774.35	779.11	0.22	1841.60	2154.56	$\leq$13.48	F	F	–	F	F	–	855.63	800.57	$\leq$6.68

TABLE II

EPFL 4-INPUT INVENTORY: PHYSICAL AREA (A), UNLOADED BLOCK DELAY (D), AND MAPPING RUNTIME IN SECONDS (T). $\leq$ DENOTES A CONSERVATIVE UPPER BOUND; – DENOTES UNAVAILABLE RUNTIME. ONE UNIFORM EGRAPHMAP PROFILE AT $\rho = 1/2$; s MARKS ITS BASIC SOURCE COVER, NOT SUCCESSFUL OPTIMIZATION. F RETAINS FAILURE STATUS.

Design	ABC map			emap			EqMap			E-Syn			E-morphic			eGraphMap		
	A	D	T	A	D	T	A	D	T	A	D	T	A	D	T	A	D	T
adder	92.42	2574.36	0.08	90.29	2569.15	0.03	189.10	6503.20	$\leq$3.08	F	F	–	84.90	2366.31	7.20	87.61	2569.15	$\leq$2.66
bar	295.84	171.90	0.13	284.48	171.90	0.06	143.49	286.49	$\leq$3.34	F	F	–	243.52	165.55	16.15	295.98	171.90	$\leq$5.77
div	3623.97	43110.43	1.71	3431.46	43091.20	1.03	F	F	–	F	F	–	F	F	–	5057.55	43093.65	$\leq$33.07
hyp	16430.32	196265.73	11.19	14772.79	195673.04	5.85	F	F	–	F	F	–	F	F	–	22414.78^s	667115.20	$\leq$65.95
log2	2015.69	3990.61	2.03	1912.04	3874.17	1.01	F	F	–	F	F	–	F	F	–	2722.24	3885.93	$\leq$33.23
max	234.67	2076.21	0.14	215.80	2075.32	0.08	337.05	5326.56	$\leq$6.38	F	F	–	218.91	1674.59	26.90	243.34	2075.32	$\leq$5.33
multiplier	1793.08	2662.96	1.32	1822.52	2631.50	0.65	F	F	–	F	F	–	F	F	–	2051.75	2742.47	$\leq$18.03
sin	420.61	1857.54	0.38	431.79	1788.13	0.19	729.73	3902.84	$\leq$11.26	F	F	–	F	F	–	559.28	1784.61	$\leq$18.93
sqrt	1633.31	47071.95	1.13	1562.46	47047.28	0.58	F	F	–	F	F	–	F	F	–	1915.41	51470.40	$\leq$28.48
square	1183.37	2510.32	0.83	1097.21	2500.72	0.46	2672.81	5939.37	$\leq$25.50	F	F	–	F	F	–	1446.47	2498.85	$\leq$21.87
arbiter	766.68	898.75	0.90	766.68	898.75	0.57	1118.07	2106.97	$\leq$20.33	F	F	–	F	F	–	761.80	899.07	$\leq$13.67
cavlc	41.31	186.01	0.06	39.83	186.01	0.03	70.39	487.44	$\leq$2.18	37.71	279.64	1.27	42.29	178.52	8.53	40.16	185.87	$\leq$2.11
ctrl	7.02	103.17	0.05	7.00	103.17	0.01	8.95	181.74	$\leq$1.55	6.80	148.26	1.08	7.47	91.38	4.03	6.75	103.17	$\leq$0.33
dec	30.83	65.72	0.05	30.83	65.72	0.03	40.55	117.96	$\leq$1.47	27.00	92.76	1.24	33.38	80.64	4.18	31.79	65.72	$\leq$1.32
i2c	72.04	165.79	0.07	70.67	165.79	0.04	94.43	391.13	$\leq$1.93	F	F	–	71.27	147.95	11.21	70.52	165.79	$\leq$4.56
int2float	13.12	181.00	0.05	13.05	181.00	0.01	19.38	296.96	$\leq$1.24	12.59	227.09	1.20	15.08	137.19	7.55	13.30	181.00	$\leq$0.47
mem_ctrl	2691.35	1093.57	1.64	2568.81	1093.57	1.12	F	F	–	F	F	–	F	F	–	3802.98	1087.09	$\leq$34.80
priority	64.17	2045.58	0.06	63.29	2045.58	0.04	53.88	3008.19	$\leq$2.92	41.89	2443.93	1.22	40.81	1071.64	19.40	64.98	2045.58	$\leq$1.89
router	12.41	187.47	0.05	14.60	187.47	0.02	10.43	540.58	$\leq$1.57	10.21	345.70	1.23	12.72	164.24	4.51	12.85	187.47	$\leq$0.37
voter	961.44	726.28	0.55	1028.35	687.07	0.34	1530.67	1531.43	$\leq$13.48	F	F	–	F	F	–	1081.87	742.90	$\leq$12.79

C. Baseline adaptations

EqMap maps six-input LUTs with original two-second per-partition, 20,000-node, 16-iteration settings, followed by same-inventory ABC amap. Area/depth profiles and controls omitting EqMap are retained. E-Syn retains frozen models, DAG features, two seconds, a soft 20,000-node threshold, 16 iterations and up to 30 ranked candidates. Its full-library calibration and upstream candidate indexing limit interpretation. E-morphic runs final native SA: five rewriting iterations, ten initial samples and six rounds of ten neighbors at probability 0.8. Feedback Liberty uses the same named cells; its loaded score differs from public delay. The reported network is the unseeded native winner.

D. Independent component removals

Table III preserves the older 20,000-node study over all 20 designs and both inventories. Its executable and resource policy differ from the proportional sweep; it does not isolate

TABLE III

HISTORICAL INDEPENDENT REMOVALS UNDER THE FIXED 20,000-NODE PROFILE, BOTH INVENTORIES; THESE ARE NOT ABLATIONS OF THE PROPORTIONAL PROFILE. A/D COUNTS ARE LOWER/EQUAL/HIGHER FOR THE ABLATED SETTING AMONG JOINTLY LEGAL FULL-SUITE PAIRS. REPEATED CEC IS OUT OF 18 ATTEMPTS PER SETTING; REPEATED A/D USE SAME-ROUND JOINTLY LEGAL PAIRS. FAILURE COVERAGE IS RETAINED, NOT FOLDED INTO PPA COUNTS. T IS THE MEDIAN MAPPING-PROCESS WALL UPPER BOUND (SECONDS) OVER SUCCESSFUL FULL-SUITE RUNS, EXCLUDING EXTERNAL PREPARATION, CEC AND EVALUATION.

Setting	CEC/40	A	D	T	Repeat CEC/18	Repeat A	Repeat D
Full configuration	34/40	–	–	$\leq$6.50	18/18	–	–
No functional quotients	34/40	16/6/12	1/25/8	$\leq$5.86	18/18	6/3/9	3/9/6
One-witness structural cuts	34/40	5/18/11	0/34/0	$\leq$6.45	18/18	5/8/5	0/18/0
No input-phase matching	22/40	11/2/9	0/2/20	$\leq$1.52	12/18	12/0/0	0/0/12
Canonical pin assignment	28/40	16/1/11	0/0/28	$\leq$1.86	12/18	12/0/0	0/0/12
No SDC matching	34/40	6/14/14	0/34/0	$\leq$5.81	18/18	4/8/6	0/18/0
No post-phase area recovery	34/40	0/2/32	0/34/0	$\leq$5.49	18/18	0/0/18	0/18/0
No seed-metric protection	34/40	3/22/9	0/34/0	$\leq$6.44	18/18	5/7/6	0/18/0

TABLE IV

PROPORTIONAL-CAPACITY SWEEP, BOTH INVENTORIES. V IS VERIFIED COVERAGE; O SELECTS AN OPTIMIZED COVER; S RETAINS THE BASIC SOURCE COVER. GEOMETRIC AREA AND DELAY RATIOS USE THE COMMON VERIFIED SCOPE AND $\rho = 1/2$ AS REFERENCE.

ρ	V	O	S	$A/A_{1/2}$	$D/D_{1/2}$
1/2	40/40	38	2	1.000000	1.000000
1	40/40	38	2	1.000095	1.000000
2	40/40	38	2	1.000980	1.000000

compaction, root retention or proportional capacity. The seven removals are independent. Functional quotients are disabled in both initial cuts and exchange. One-witness cuts restrict optional structural enumeration while preserving mandatory source fallbacks. Input-phase removal rejects complemented assignments but retains explicit inverters. Canonical pin assignment restricts assignment freedom, not pin-delay lookup alone. SDC removal sets the boundary-support allowance to zero without introducing unions.

Area-recovery removal omits post-phase required-time flow and actual-area substitution; seed initialization remains. Incumbent removal disables the dual-metric envelope, seed-area ceiling and bound-feasible DA cache, while ordinary selection, cycle checks and generator proofs remain. Neither removal changes correctness checks. All settings retain the same node, cut, pool and total-time limits. No replacement stage is added, although earlier completion changes the time available to later stages.

Three rounds use sqrt, sin and priority in both inventories with rotated setting order and alternating circuit/library order. Per-case values, failures and ranges remain archived. Paired counts require two valid covers; coverage is separate. These targeted results do not establish full-suite stability or additive component effects.

VI. DISCUSSION AND CONCLUSION

The capacity split permits complete physical source covers while preserving separate search allowance. A verified fallback establishes representability, not optimization success. Additional nominal capacity does not improve delay when admissions are unchanged or another stage dominates.

Relative to emap, sqrt changes area/delay by -3.56%/-0.043% in K3 (area 1586.26, delay 60061.59) and +22.59%/+9.401% in K4 (area 1915.41, delay 51470.40). These fixed-configuration observations are not replaced by historical winners or the best repetition.

Div and mem_ctrl select optimized covers in all six repetitions per design. Hyp instead retains verified source covers after the unchanged 60-second native guard. K4 stops during cut propagation; K3 completes approximately 25.7 seconds of saturation, 15.4 seconds of generation and 9.7 seconds of alias normalization before stopping during pruning.

The targeted study adds 54 attempts on sqrt, sin and bar (three ratios, two inventories, three rounds) and 18 on div, hyp and mem_ctrl at the selected ratio. Fresh optimized-output checks and reused source CEC proofs are identified separately. These limited repeats retain failures and regressions; they establish neither machine-independent nor full-suite stability.

The K6 extension covers 40 of 47 single-output cells, excluding seven seven-to-nine-input gates. All 120 baseline groups were attempted. Independent CEC validates 190 of 260 method observations, including four verification-only hyp supplements; 70 remain failed. Earlier K3/K4 tables are not a same-binary sensitivity study.

Earlier matched-budget recovery/support studies used heterogeneous per-design search ratios, not a uniform final configuration. Neither improves delay, and both retain area regressions.

Uniform-$\rho = 0.5$ sqrt retains delay 50815.83 and area 2018.65, versus emap's 46826.06 and 1540.99. Output-root early termination leaves sqrt and sin unchanged. Additional original boundaries (2048 to 7075 helpers) yield an internal root delay of 46799.83, but allocation failure or the 60-second native timeout prevents an optimized export. Exact normalization and ranking reuse remove duplicated work but have not resolved pool-retention costs. Final CEC validates only source covers in these failed trials. Full-suite repetition and optimized hyp mapping remain pending.

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